

Test I Solutions
CS4033E Quantum Computation

1. Let U be a single qubit gate whose matrix is Hermitian. Show that for any state $|\phi\rangle$, $(U + I)|\phi\rangle$ is an Eigen vector of U .

Soln: U must be Unitary as it is a gate, and is Hermitian by assumption. Hence $U^*U = U^2 = I$. That is, $(U - I)(U + I)|\phi\rangle = 0$ for all $|\phi\rangle$. Let $|\psi\rangle = (U + I)|\phi\rangle$. Then we have $(U - I)|\psi\rangle = 0$ or $U|\psi\rangle = |\psi\rangle$. Thus $|\psi\rangle$ is an Eigen vector of U with Eigen value $+1$.

2. Write down the single 4×4 two qubit transformation corresponding to applying the Hadamard gate on the second qubit only, in a two qubit system.

Soln: Direct question. By the superposition postulate of QM, the required matrix is $I_2 \otimes H$, where I_2 is the 2×2 identity matrix and H is the Hadamard matrix $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. (Note: Here, the question is a direct one - asking you to compute the matrix, with no further requirements. In such cases, you need to calculate the full 4×4 matrix, and not just leave the abstract expression $I_2 \otimes H$ for full credits.)

3. Which among the two bit quantum states is a superposition of two single qubit states? $|\phi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$, $|\phi_2\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, $|\phi_3\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$, $|\phi_4\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)$. In each case, either show the product representation, or prove that no such representation is possible.

Soln: $|\phi_1\rangle$ can be decomposed into a superposition of two single qubit states: $|\phi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. On the other hand, if $|\phi_2\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$, Then by comparing coefficients, we have $\alpha\gamma = 1 = \beta\delta$, which requires $\alpha, \beta, \gamma, \delta$ to be non-zero. This is contradictory to the requirement $\alpha\delta = 0 = \beta\gamma$. The case of $|\phi_3\rangle$ and $|\phi_4\rangle$ are handled similarly.

A state which cannot be decomposed into a tensor product of single qubit states is said to be in an *entangled state*.

4. Write down the 4×4 Hermitian matrix M that corresponds to the measurement of the second bit of a 2 qubit quantum system. Assume that the real valued measured outcomes are ± 2 corresponding to $|0\rangle$ and $|1\rangle$ observed on the second bit.
- Write down the spectral decomposition of M . (That is, express M as a linear combination of projection operators.) You must give clear justification for the answer.
 - When each of the four ket vectors defined in the previous question - $|\phi_1\rangle, |\phi_2\rangle, |\phi_3\rangle, |\phi_4\rangle$ - are measured using M . What are the possible post measurement states of the system. What are the probabilities of each of these post measurement states?

Soln: From the given data, it is easy to analyse M based on the action on the standard basis states. $M|00\rangle = 2|00\rangle$, $M|01\rangle = -2|01\rangle$, $M|10\rangle = 2|10\rangle$, and $M|11\rangle = -2|11\rangle$. This shows that for this particular matrix M , the standard basis states themselves form four orthogonal Eigen vectors, and the data in the question specifies their corresponding Eigen values. Hence, by spectral theorem, M is given by:

$$M = 2(|00\rangle\langle 00| + |10\rangle\langle 10|) - 2(|01\rangle\langle 01| + |11\rangle\langle 11|). \quad (1)$$

This solves Sub question [a]. If you substitute vectors for $|00\rangle, |01\rangle, |10\rangle$ and $|11\rangle$ in Equation 1, you will get the matrix M required to be computed in the main question. (If your arguments leading to Equation 1 are very precise, even if you don't expand the symbolic expression to get a full 4×4 matrix, you will get full credits!)

For Sub question [b], note that the projection operators corresponding to the two Eigen values $+2$ and -2 are:

$$P_0 = |00\rangle\langle 00| + |10\rangle\langle 10| \quad (2)$$

$$P_1 = |01\rangle\langle 01| + |11\rangle\langle 11| \quad (3)$$

CS6101E Problem Set 1

For $|\phi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$, by the measurement postulate, the probability of obtaining measurement outcome +2 (corresponding to observing the second qubit in state $|0\rangle$) is $\|P_0|\phi_1\rangle\|^2 = \|\frac{1}{\sqrt{2}}|00\rangle\|^2 = \frac{1}{2}$. Similarly, the probability of obtaining measurement outcomes -2 (which corresponds to observing the second qubit in state $|1\rangle$) is $\frac{1}{2}$. The cases of $|\phi_2\rangle, |\phi_3\rangle, |\phi_4\rangle$ are handled similarly.

5. Consider the 2 qubit gates G, U with the following functionality: $G|00\rangle = |00\rangle, G|01\rangle = |01\rangle, G|10\rangle = |11\rangle$ and $G|11\rangle = |10\rangle$; $U|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle), U|01\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle), U|10\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle), U|11\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle)$. Consider the quantum circuit that on input $|10\rangle$, first applies the gate U and then applies G . Suppose the first qubit of the output is measured, what are the various post measurement states and the probabilities of each of the post measurement states?

Soln: This is straightforward. $U|10\rangle$ leaves the system in the state $\frac{1}{\sqrt{2}}(|00\rangle - |10\rangle)$. G applied to this state leaves the system in the state $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$. It is clear that in this state, the probability of measuring the first qubit to $|0\rangle$ is the same as the probability of measuring the first qubit to $|1\rangle$, and both must be $\frac{1}{2}$.

An interesting property of this state is that measurement of the state of either of the two qubits (with respect to the standard basis) will fix the state of the other qubit as well. On the other hand, before any measurement is done, they are in a superposition of $|00\rangle$ and $|11\rangle$ states.